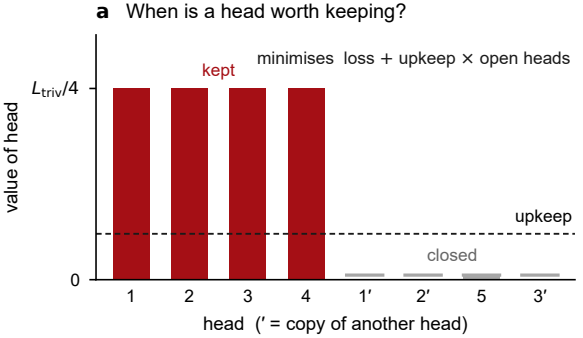
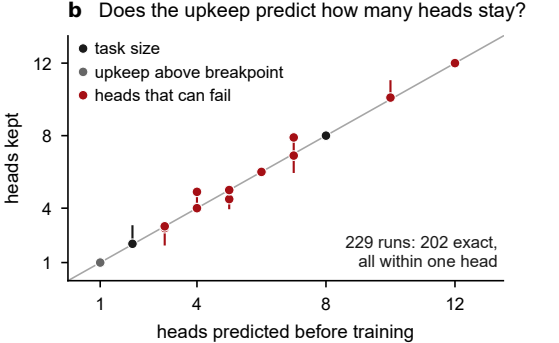
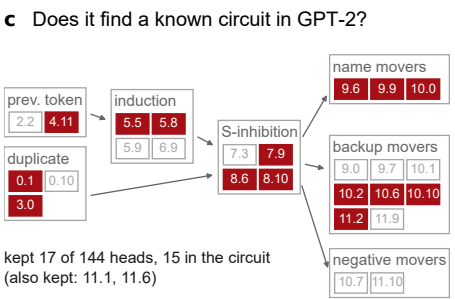
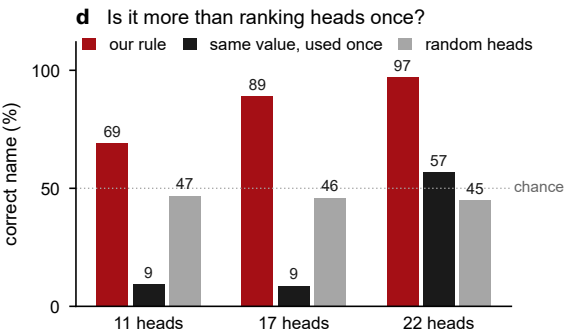

a When is a head worth keeping?
minimises loss + upkeep × open heads
kept
$L_{triv}/4$
value of head
upkeep
closed
0
1
2
3
4
1′
2′
5
3′
head (′ = copy of another head)


b Does the upkeep predict how many heads stay?
task size
upkeep above breakpoint
heads that can fail
heads kept
12
8
4
1
229 runs: 202 exact,
all within one head
1
4
8
12
heads predicted before training


c Does it find a known circuit in GPT-2?
prev. token
2.2
4.11
induction
5.5
5.8
5.9
6.9
duplicate
0.1
0.10
3.0
S-inhibition
7.3
7.9
8.6
8.10
name movers
9.6
9.9
10.0
backup movers
9.0
9.7
10.1
10.2
10.6
10.10
11.2
11.9
negative movers
10.7
11.10
kept 17 of 144 heads, 15 in the circuit
(also kept: 11.1, 11.6)


d Is it more than ranking heads once?
our rule
same value, used once
random heads
correct name (%)
100
50
0
69
9
47
89
9
46
97
57
45
chance
11 heads
17 heads
22 heads